

Late Winter Activity of Bats At Skaha Bluffs

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ABSTRACT

Winter bat activity in British Columbia has been largely overlooked. Only one publication has been written (Nagorsen et al. 1993). This paper summarized winter bat observations in the province, which consisted of observations of winter bats at a variety of hibernacula features. As such, much of our understanding of winter bat ecology is based on information gained elsewhere, and assumed to apply to our population of bats in BC. Studies being done in a number of areas in BC are showing that our assumptions appear to be wrong in some cases.

Acoustic monitoring at Skaha Bluffs revealed extensive winter bat activity with a total number of bat passes of 1551 over 105 sample nights. Of the ten sites that were sampled, average number of bat passes per night were quite variable, ranging from 2 to 59. At least eleven species were recorded during the sampling period (07 December 2011 to 31 March 2012) and therefore are thought to be hibernating in the area. Some species seemed to be more active than others and some were only detected near the end of the sampling period when emergence was suspected to occur. Not all species residing in the Okanagan region during summer were detected in winter, suggesting that some species either migrate out of the area or are not active during winter, perhaps remaining in a deeper state of hibernation, or choosing different types of hibernacula. Monitoring in the 'shoulder seasons' of spring and fall may elucidate whether these species are hibernating in the area or migrate out.

Notable detections included little brown myotis (Endangered), pallid bats (Threatened), spotted bat (Special Concern), Townsend's big-eared bat (provincially Blue-listed), western small-footed myotis (provincially Blue-listed) and possibly canyon bat (not documented for the province yet).

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INTRODUCTION

The general area referred to as Skaha Bluffs includes private lands, provincial park and conservation lands jointly held by the Nature Conservancy of Canada, The Land Conservancy of BC and BC Parks. The landscape is dominated by steep gneiss cliffs, which have become a mecca for rock climbers and the primary management attention by BC Parks. Past reports have raised concerns about the affects of rock climbing on the environment (Richardson 1994, Chapman 1995, Sarell 1995, Urban Systems 1995, McConnel 1996). The development of new climbs has not been managed and this could be causing the destruction or disruption of summer and winter bat roosts.

The federally listed pallid bat (*Antrozous pallidus*), resides in the Okanagan region of BC. The spotted bat (*Euderma maculatum*) , is also of federal interest, and is rare but found in areas of cliff habitat in south-central BC. It was unknown whether these species remain in BC during the winter to hibernate in situ, or if they migrate. Winter ecology of these species is unknown, as is winter habitats. In other areas there are records of each of these species being active in winter (D. Johnston, California, pers. comm.; P. Ormsbee, USFS, Oregon, pers. comm.).

As White Nose Syndrome is a threat to bats during winter, and is a deadly disease currently in the eastern part of Canada and is spreading west (USFWS WNS 2012), learning about the winter ecology of our federally and provincially listed bat species is prudent.

This study focused on bat activity in winter as it is being found that some species are quite active during the winter and this might indicate some of the bat species that are hibernating at Skaha Bluffs. Additionally, it contributes to establishing a baseline of bat activity so that the effects of ongoing rock climbing might be gauged as well as the effects of White Nose Syndrome (WNS) when it ultimately becomes established in BC. The goal was to place many of the detectors as close to potential winter roosts as possible to try to maximize the chance of recording bats at their hibernacula.

METHODS

Bat detectors were used to record echolocation calls (and other vocalizations) produced by active bats during the winter of 2011/2012. Each site used an SM2 bat detector produced by Wildlife Acoustics, which records in full spectrum with the exception of detector named Oph5 which was an SM2Bat+. This new model of detector had the option of recording in Zero Crossing Analysis mode which uses less memory and less power. Detectors store data on memory cards, which can be switched out and downloaded to a computer for analysis.

Full spectrum data recorded on the SM2 bat detectors were run through a program called Sonobat (version 3.05 Great Basin) which automatically assigns a species identity, or at least a grouping of which bat species the program determines it to be, based on a 90% statistical discrimination. Species are assigned a category only if they fall close enough to the statistical centroid of one species, and otherwise they are given a mixed classification. This software provides the caveat that a 90% probability of discrimination means it falls near this centroid and does not guarantee that the species has been classified correctly given that some species have large overlap of call features with other species, and their centroids may be relatively close (J. Szewczak, Sonobat, Humboldt University, California). Echolocation calls that were recorded using Zero Crossing Analysis (Oph5) were visually analyzed using Analook software to determine which species, or species group, generated the call, based on characteristic features, typical for that species or species group.

Sites were only sampled for one to two weeks so rechargeable batteries were used to power the units and were changed each time the detector was moved to a new site.

Ten sites were surveyed for bat activity at Skaha Bluffs (Figure 1, Table 1). Each site was sampled for a minimum of seven nights to improve the likelihood that at least one of the nights would be favourable for some bats to emerge from hibernacula in the area and be recorded echolocating in flight. All sites were selected for the potential to have hibernacula in the immediate vicinity. Hibernacula were assumed to occur in deeply fractured bedrock or blocky colluvium, which would enable bats to get beyond the frost line. One site, Cave Hill, has large enough fractures to allow human access to some of the fractures. Only one of the sites (Oph5C No Climbs) did not have a developed climb on it and was thought to have little public use. The first recordings began on 07 December, 2011 and the last recording was 31 March, 2012, for a total of 105 sampling nights. A sampling night was considered to be at least four hours of recordings after sunset. Other studies have shown that this is the most active period during the winter. Full nights were not recorded due to the power consumption and need for more frequent visits to replace batteries.

Table 1: Detection sites for bat activity at Skaha Bluffs

Detector Name	Site Name	Z	UTM_E	UTM_N	Elev	Start Date	End Date	Nights
Oph3D	Fern Gully	11	314336	5480648	640	9-Mar	20-Mar	12
Oph5A	Fortress	11	313861	5479214	538	27-Feb	5-Mar	8
Oph5B	Wave	11	314031	5479365	569	6-Mar	12-Mar	7
Oph5C	No Climbs	11	314217	5479418	587	13-Mar	19-Mar	7
Oph5D	Great White	11	314281	5479695	622	20-Mar	27-Mar	8
Oph6A	Whitewash	11	313922	5478954	544	27-Feb	5-Mar	8
Oph6B	Unknown	11	314205	5479048	556	6-Mar	12-Mar	7
Oph6C	Blazing	11	314194	5479562	588	13-Mar	19-Mar	7
Oph6D	Go Anywhere	11	314286	5479909	614	20-Mar	27-Mar	8
Sum1A	Cave Hill	11	314366	5480531	662	7-Dec	4-Jan	22
Oph3E	Cave Hill	11	314366	5480531	662	22-Mar	31-Mar	11
Total								105

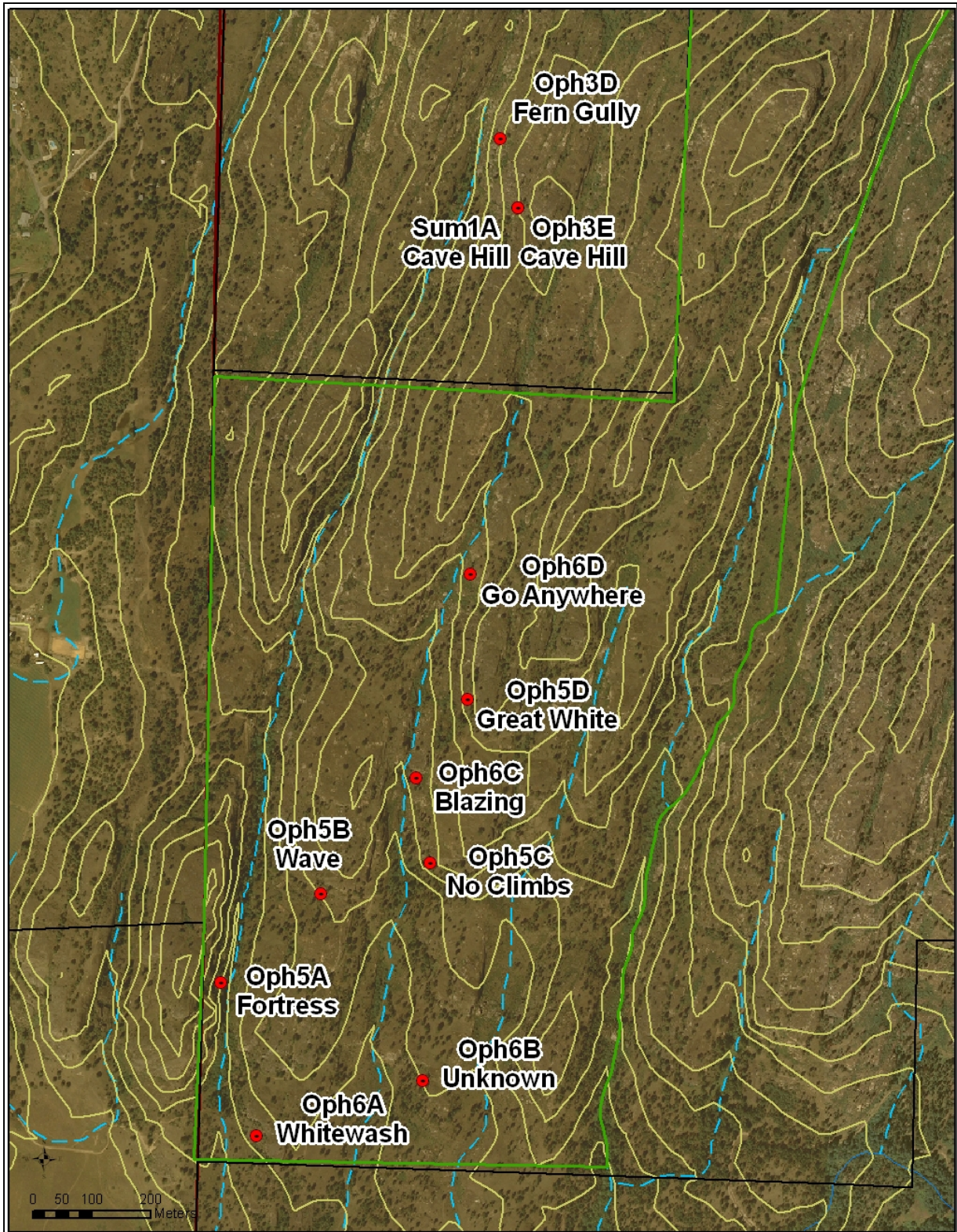


Figure 1: Locations of bat detector sites at Skaha Bluffs.

RESULTS

A total of 1551 passes were recorded at the ten sites over a total of 105 sample nights at Skaha Bluffs. Overall, there was an average rate of 14.8 bat passes per night. This is very much greater than the average level of activity found at other sites in the Okanagan and Thompson (Sarell et al. 2012). However, the volume of bat activity varied greatly between sites, ranging from an average of 2.0 passes per night to 55 passes per night. Two passes/night is about how many were being recorded at mine sites in the Thompson and Okanagan (Sarell et al. 2012). One site at Skaha Bluffs (Oph6a) yielded an average of 55 passes per night and during one of those nights (04 March 2012) there were 277 passes. Two other sites (Oph6B and Oph3E) also yielded more than 100 passes in a single night. The high degree of variation in volumes of bat activity between sites suggests that bat activity is localized and likely concentrated around hibernacula.

Table 4: Volume of bat activity at detector sites at Skaha Bluffs.

Detector Name	Site Name	Start Date	End Date	Nights	Passes	Ave Pass /Night	Max Pass /Night
Oph3D	Fern Gully	9-Mar	20-Mar	12	109	9.1	47
Oph5A	Fortress	27-Feb	5-Mar	8	116	14.5	69
Oph5B	Wave	6-Mar	12-Mar	7	31	4.4	16
Oph5C	No Climbs	13-Mar	19-Mar	7	71	10.1	54
Oph5D	Great White	20-Mar	27-Mar	8	36	4.5	10
Oph6A	Whitewash	27-Feb	5-Mar	8	439	54.9	274
Oph6B	Unknown	6-Mar	12-Mar	7	198	28.3	122
Oph6C	Blazing	13-Mar	19-Mar	7	14	2.0	8
Oph6D	Go Anywhere	20-Mar	27-Mar	8	19	2.4	11
Sum1A	Cave Hill	7-Dec	4-Jan	22	100	4.5	19
Oph3E	Cave Hill	22-Mar	31-Mar	11	418	38.0	177
Total				105	1551	14.8	

The most ubiquitous species of bats detected during the winter sampling period were California myotis (MYCA), silver-haired Bat (LANO), and big brown bat (EPFU) (Table 3). California myotis have been reported to be very active during winters in Nevada (O'Farrell et al. 1967) and big brown bats have been known to hibernate in sites with only modest insular protection (Nagorsen et al. 1993). Lausen et al. (2006) found big brown bats to be highly active in winter on the Prairies.

Table 2: Legend of species acronyms and acoustic categories.

Species/Category	Description
ANPA	<i>Antrozous pallidus</i>
COTO	<i>Corynorhinus townsendii</i>
EPFU ¹	<i>Eptesicus fuscus</i>
LANO ¹	<i>Lasionycteris noctivagans</i>
Low F	one of the following possible species: COTO, EPFU, LANO
High F	one of the following possible species: PAHE, or a <i>Myotis</i> spp.
MYCA	<i>Myotis californicus</i>
MYCA/MYYU	<i>M. californicus/yumanensis</i>
MYCA/PAHE	<i>M. californicus/Parastrellus hesperus</i>
MYLU/MYCI/MYVO	<i>M. lucifugus/ciliolabrum/volans</i>
MYCI	<i>M. ciliolabrum</i>
MYLU	<i>M. lucifugus</i>
MYVO	<i>M. volans</i>
<i>Myotis</i> spp.	one of the following possible species: MYCA, MYYU, MYEV (<i>M. evotis</i>), MYVO, MYLU, MYCI
MYYU	<i>M. yumanensis</i>

Table 3: Species detections at locations Skaha Bluffs.

Species	Sum1 Cave Hill	Oph3E Cave Hill	Oph3d Fern Valley	Oph5a* Fortress	Oph5b* Wave	Oph5c* No Climbs	Oph5d* Great White	Oph6a White wash	Oph6b Un known	Oph6c Blazing	Oph6d Go Anywhere
ANPA	1	21	2	1	0	0	0	0	0	0	4
COTO	1	7	1	0	0	0	0	0	0	0	0
EPFU ¹	3	29	12	32	0	2	0	12	2	1	1
EUMA	0	1	0	0	0	0	0	0	0	0	0
LANO ¹	8	28	7	4	0	0	0	8	0	0	2
Low F	15	0	19	0	0	0	0	0	0	0	0
High F	0	1	0	0	0	0	0	0	0	0	0
MYCA	15	128	4	0	0	0	0	417	188	11	10
MYCA/MYYU	0	0	0	57	20	54	21	0	0	0	0
MYCA/PAHE	0	20	0	4	0	7	1	1	4	0	2
MYLU/MYCI/MYVO	0	0	0	7	5	6	8	0	0	0	0
MYCI	7	32	0	0	0	0	0	0	0	0	0
MYEV	0	5	0	0	0	0	0	0	0	0	0
MYLU	1	51	8	0	0	0	0	0	3	2	0
MYVO	0	27	0	0	0	0	0	0	0	0	0
<i>Myotis</i>	0	0	0	10	6	2	6	0	0	0	0
MYYU	2	5	4	0	0	0	0	0	0	0	0

* These sites (all Oph5 detectors) were recorded using Zero Cross Analysis unlike the remaining sites that used full spectrum to record calls.

¹ These 2 species (EPFU and LANO) are acoustically very similar, and therefore some misidentification should be assumed.

DISCUSSION

This acoustic monitoring revealed a tremendous amount of bat activity taking place at Skaha Bluff. Ten sites were monitored for at least a seven-night period and 1551 bat passes were detected during the 105 nights that were sampled. At least 11 species were detected, with some species demonstrating more activity than others. High levels of activity may be a result of large populations of a species hibernating at a site or just very high levels of activity of a few individuals. Nights with high levels of activity were localized, suggesting that bats do not travel very far when they are active. This may be due to being detected at some point, although not all were equally active. High levels of activity that were found at most of these sites provides strong evidence that hibernacula likely are located near the sites.

The following itemizes species of interest:

PALLID BAT

Pallid bats are known only to occur in the South Okanagan and Lower Similkameen Valley in Canada (The Pallid Bat Recovery Team 2008). They were first discovered in Canada in 1931 (Racey 1933). Two more recent intensive surveys using radiotelemetry has expanded out knowledge of the ecology of pallid bats in BC (Chapman et al. 1994, Rambaldini 2006). No overwintering records have been reported in BC (Nagorsen et al. 1993). Prior to this survey, the latest pallid bat detection in BC was 25 October 2006 when overnight temperatures were dropping to -8°C (Sarell et al. 2007). This led researchers to suspect that pallid bats were overwintering in BC. This survey clearly demonstrates that pallid bats are hibernating in BC and have low levels of activity, as has been demonstrated in Nevada (O'Farrell et al. 1967).

Pallid bats were detected on 29 occasions at 5 sites. These Skaha Bluff sites are outside the previously known range extent for this species. Mist netting would help to confirm their presence, as the calls of pallid bats are similar to clutter calls of big brown bats. Cave Hill, outside the Provincial Park, had the most detections.

TOWNSEND'S BIG-EARED BAT

Townsend big-eared bat was detected at three of the ten sites. The most prolific site was Cave Hill, which had seven passes. Two Townsend's big-eared bats were observed hibernating in the large fractured bedrock at Cave Hill on 13 November 2011 but were not there during a visual survey on 07 December 2011. This species seems to be readily visible in mines during hibernation and is active to some extent, switching roosts and flying mid-winter. This has been found to be the case in the West Kootenay also (T. Hill, FWCP, pers. comm.; C. Lausen, pers. obs.).

SPOTTED BAT

Spotted bats have been documented using the cliffs as summer roosts (Sarell et al. 1996). There was only one detection of a spotted bat during the sampling period and it was in late March so this suggests that spotted bats are not active during the cold portion of winter. This is consistent with other regional findings (Sarell et al. 2012). The spotted bat was detected in the vicinity of Cave Hill (Oph3e).

LONG-EARED BATS

It appears that the long-eared bat species (*M. thysanodes* and *M. evotis*) are not active during the winter. Only one Western long-eared bat was detected and that was on the 25 March 2012 at Cave Hill. This is consistent with the results found in the rest of the Okanagan and Similkameen (Sarell et al. 2012). The fringed myotis (*M. thysanodes*) has been captured at Skaha Bluffs climbing area (Sarell et al. 1996) and at Gilles Creek (Sarell 2008).

LITTLE BROWN MYOTIS (AND OTHER 40 KHz MYOTIS)

The little brown myotis (*M. lucifugus*) was most abundant towards the end of sampling, suggesting that they go into deep hibernation and rarely arouse during the winter. This is consistent with what was found in the other parts of the Okanagan and Thompson (Sarell et al. 2012). The little brown myotis has recently been listed by COSEWIC as Endangered in Canada. This species, together with other colonial roosting myotis, are likely to be at high risk from White Nose Syndrome. Western small-footed myotis (*M. ciliolabrum*) were only detected at Cave Hill, despite Skaha Bluffs being known to provide summer roosts in the rock crevices (Sarell et al. 1996). The long-legged myotis (*M. volans*) also was found only at Cave Hill where it appears to have hibernated.

CALIFORNIA MYOTIS (AND OTHER 50 KHz MYOTIS)

Californian myotis was the most active myotis species in winter, which is consistent with observations in BC (Sarell et al. 2012) and other parts of their range (O'Farrell et al. 1967). F50 kHz myotis (*M. californicus* and *M. yumanensis*) were detected at all of the 10 sites sampled. This suggests they are either hibernating in most of these rock faces and/or they do not restrict their flight to the vicinity of their roosts.

CANYON BAT

Canyon bats have not formally been documented in BC but there is strong evidence to believe they occur here, based on echolocation call recordings and two captures of individuals. All encounters have been in the South Okanagan and Lower Similkameen. If this species was verified as occurring in Canada, it would likely be listed as Endangered or Threatened by COSEWIC due to restricted range and low population levels. Canyon Bats are known to be active during winters in Nevada (O'Farrell et al. 1967) so one would expect them to be detected in the South Okanagan if they were resident bats. Unfortunately neither Sonobat nor Analook were able to discriminate between 39 echolocation calls made at seven sites that could have been either Canyon Bat or California Myotis.

SUMMARY

Skaha Bluffs is one of the most active winter bat areas known in the province. Some of these bats may be resident at Skaha Bluffs, but it certainly appears that the abundance of deep fractures in the cliff faces attract bats from a much broader area. It is also likely that these hibernacula have been used by countless generations of bats as bats are long lived (> 40 years) and have high site fidelity to roosts.

There was some recreational climbing during the sampling period, as there was likely climbing activity when bats were entering hibernation. The data collected during this study cannot ascertain what affects climbing activity has on hibernating bats as there is no baseline data from before climbing began at Skaha Bluffs. But it is generally supported that frequent arousal can be extremely detrimental as important fat stores are metabolized, leading to starvation. This may be extremely important for bat species that do not normally arouse during hibernation (ie. long-eared bats, 40 kHz bats, spotted bats).

As White Nose Syndrome continues to spread west it is likely that BC's bat species will eventually be affected. Understanding winter bat ecology prior to potential mass mortality would be prudent in the event that potential mitigation strategies become available. Identifying what species of bats overwinter at Skaha Bluffs and determining habitat requirements for winter hibernacula for each species would provide valuable data for appropriate management of winter bat habitat. Establishing such baseline data will require further winter research: acoustic monitoring, together with mistnet capture and radiotelemetry.

RECOMMENDATIONS

1. Expand acoustic detections to all cliffs at Skaha Bluffs during the spring and fall to obtain baseline data for each cliff. This is important information for ranking which cliffs are most important for hibernating bats and also monitoring population declines when WNS becomes established.
2. Use mist netting and telemetry to confirm which species of bats overwinter at Skaha Bluffs and determining habitat requirements for winter hibernacula for each species.
3. Use the data collected to determine the hibernating period for bats at Skaha Bluffs and close the Park to climbing during this period until it can be demonstrated that climbing does not have an affect on hibernating bats.
4. Conduct summer bat surveys to establish which species of bats are roosting at Skaha Bluffs and whether they are tolerant of climbing activities.

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